Module 3 Official

April 2, 2026

0.1 Module 3: Fibrosis

0.2 Team Members: Roy Chen and Will Marschall

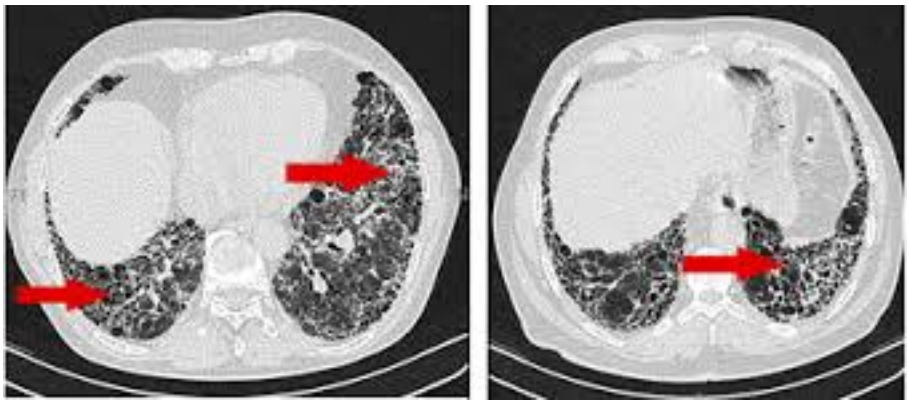
0.3 Project Title: Spatial Distribution of Fibrosis in the Murine Bleomycin Model of Lung Fibrotic Disease

0.4 Project Goal:

This project seeks to develop an image-analysis pipeline that quantifies fibrosis in black-and-white mouse lung sections by measuring the percent of white pixels at known depths into the lung, and then uses interpolation to estimate fibrosis severity at depths where a direct image measurement is not available.

0.5 Disease Background:

Pulmonary fibrosis is a chronic interstitial lung disease in which the lung tissue becomes thickened, stiff, and scarred, reducing how efficiently oxygen moves from the alveoli into the bloodstream. Idiopathic pulmonary fibrosis (IPF) is the most common idiopathic interstitial pneumonia and is a progressive fibrotic lung disease of unknown cause. As fibrosis worsens, the lungs become less compliant, gas exchange becomes less effective, and patients can develop worsening shortness of breath and eventual respiratory failure [1], [3].



(image from the Association pulmonaire du Québec)

The white speckled regions represent dense scar tissue, while darker areas represent normal lung tissue, The white specks in the image depict scar tissue replacing normal lung tissue, which reduces the amount of functional area available blood to be oxgyenated

0.6 Prevalence and Incidence:

Pulmonary fibrosis is relatively uncommon, but it remains a serious and clinically important disease. The Pulmonary Fibrosis Foundation estimates that more than 250,000 Americans are living with pulmonary fibrosis and that approximately 50,000 new cases are diagnosed each year. One recent estimate cited by the Pulmonary Fibrosis Foundation suggests that idiopathic pulmonary fibrosis affects about 1 in 200 adults over the age of 70 in the United States. NHLBI also notes that IPF is diagnosed most often in adults in their 60s or 70s and is more common in men than women [2], [3].

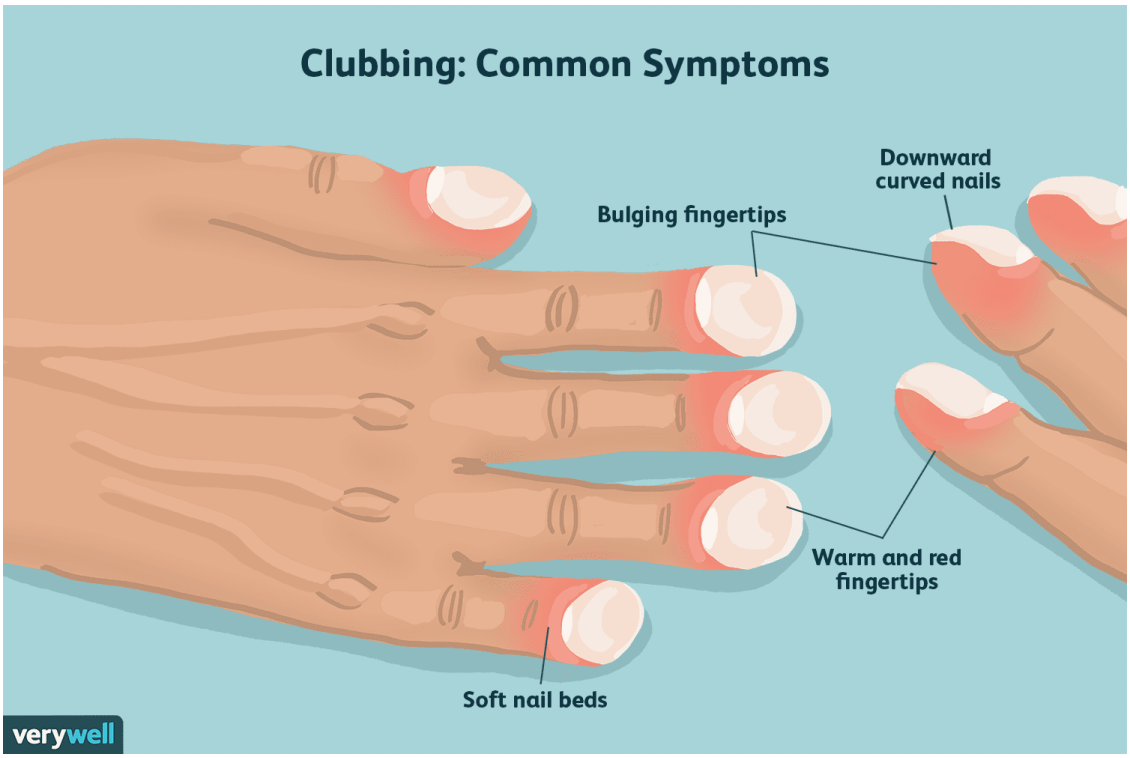
(image provided by Healthline)

0.7 Risk Factors:

Although the exact cause of IPF is unknown, several factors are associated with increased risk. Major risk factors include older age, smoking, male sex, family history, and genetic susceptibility. NHLBI notes that mutations in multiple genes have been linked to increased IPF risk, and the disease is more likely in individuals with a first-degree relative who has IPF. More broadly, pulmonary fibrosis can also occur secondary to environmental or occupational exposures, certain medications, radiation, or autoimmune disease, even though those cases are not classified as idiopathic [2], [3].

0.8 Symptoms:

The most common symptoms of IPF are progressive shortness of breath and a dry cough. Other symptoms may include fatigue, gradual unintended weight loss, rapid shallow breathing, aching muscles or joints, and clubbing of the fingers or toes. NHLBI emphasizes that symptoms may develop gradually and worsen over time, and some patients may not notice symptoms early in the disease course [1], [4].



(image from Verywell health)

0.9 Standard of care treatments:

There is currently no cure for IPF, so treatment focuses on slowing disease progression, improving symptoms, and preserving quality of life. NHLBI lists the antifibrotic medications nintedanib and pirfenidone as current medical therapies that can slow lung-function decline and may reduce acute exacerbations. Other supportive treatments include pulmonary rehabilitation, oxygen therapy, and ventilatory support when needed. For some patients with advanced disease, lung transplantation may be an option, although transplantation carries substantial risks such as infection and graft rejection [5].

0.10 Biological mechanisms (anatomy, organ physiology, cell & molecular physiology):

Pulmonary fibrosis develops when repeated epithelial injury and abnormal wound-healing responses lead to excessive activation of fibroblasts and differentiation into myofibroblasts. These myofibroblasts produce excess extracellular matrix, including collagen, which thickens and stiffens the lung tissue. A central pathway in this process is TGF- signaling, which promotes fibroblast activation, myofibroblast differentiation, and extracellular matrix deposition. As scar tissue accumulates around the alveoli, the diffusion distance for oxygen increases and lung compliance decreases, making breathing more difficult and impairing gas exchange [3], [6].

0.11 Clinical Course and Prognosis:

IPF is a progressive disease, but the rate of progression varies between patients. NHLBI notes that some people decline slowly, whereas others worsen more rapidly or experience acute exacerbations.

Important complications include pulmonary hypertension and respiratory failure, both of which reflect the lung's declining ability to oxygenate the body adequately. This progressive nature is one reason why earlier detection and more accurate monitoring tools are clinically important [1], [5].

0.12 Data-Set:

Unpublished data were collected by the **Peirce-Cottler Lab (Department of Biomedical Engineering)** and the **Kim Lab (Division of Pulmonary and Critical Care)** at the University of Virginia School of Medicine [7]–[9]. Dr. John Kim is a UVA pulmonary and critical care physician-scientist whose research interests include interstitial lung disease and pulmonary fibrosis [7].

The full class dataset consisted of **78 black-and-white lung images** obtained at known depths throughout a fibrotic mouse lung [8]. The 1st lecture described a **murine bleomycin-induced lung injury model** in which mouse lungs were harvested approximately 3 weeks after injury, sectioned in the transverse plane, immunostained, imaged by microscopy, and converted into black-and-white masks for analysis [8]. In these processed images, **white regions represented fibrotic lesion area and black regions represented healthier lung tissue** [8].

The depth of each section was measured in **microns (μm)**, spanning approximately **0 to 10,000 μm** from the top to the bottom of the lung according to the initial dataset description provided [8], [9]. For our project analysis, we selected **15 images** from the provided image set and matched each filename to its corresponding depth using the provided lookup file, `Filenames and Depths for Students.csv` [9]. We then used Python to calculate the percentage of white pixels in each selected image and treated that value as a **fibrosis proxy** for the depth at which the image was obtained.

0.13 Data Analysis

0.13.1 2a. Methods

For the data analysis portion of this project, we used Python to analyze a selected subset of black-and-white mouse lung images collected at known depths into a fibrotic lung. Following the workflow from lecture and the provided `main_example.py`, each image was read in grayscale, thresholded at a pixel value of 127 to create a binary image, and then used to count white and black pixels. In these processed images, white pixels were treated as a proxy for fibrotic lesion area and black pixels were treated as healthy lung tissue.

For each image, we calculated the fibrosis proxy as the percent of white pixels in the image:

$$\text{Percent white pixels} = 100 \cdot \frac{\text{white pixels}}{\text{total pixels}}.$$

We then wrote the analyzed image data to a `.csv` file, plotted measured percent white pixels versus depth into the lung, and used both linear and quadratic interpolation to estimate the fibrosis proxy at an unmeasured depth of 345 μm. To stay consistent with lecture and `interpolation_example.py`, we also compared the built-in SciPy interpolation methods to manual matrix-based polynomial interpolation.

```
[ ]:
```

```python
# The following code is ran from our 'main.py' file, which contains the main␣
↪logic for our fibrosis analysis. You can run this code to execute the full␣
↪analysis pipeline, or you can run individual cells to test specific␣
↪functions or steps in the process.
# Our full code commenting and logic is in 'main.py'

# This cell sets up our environment and imports necessary libraries and␣
↪functions. It also defines the target depth for our analysis, which is 345␣
↪micrometers in this case.
from pathlib import Path
import sys
import importlib
import pandas as pd
from IPython.display import display

# Set pandas to display floats with 6 decimal places for better readability in␣
↪our results.
pd.options.display.float_format = "{:.6f}".format

# Define paths to our project root and code directory, and ensure that the code␣
↪directory is in our system path for imports.
PROJECT_ROOT = Path.cwd()
CODE_DIR = PROJECT_ROOT / "code"

# If the code directory is not already in the system path, add it so we can␣
↪import our main module.
if str(CODE_DIR) not in sys.path:
    sys.path.append(str(CODE_DIR))

# Import our main module, which contains all the functions and logic for our␣
↪analysis. We also reload it to ensure we have the latest version after any␣
↪edits.
import main
importlib.reload(main)

# Define the target depth in micrometers for our analysis. This is the depth at␣
↪which we will be evaluating our fibrosis predictions and measurements.
TARGET_DEPTH_UM = 345.0
```

0.13.2 2b. Analyze the selected images and create the measured data table

We first analyzed the selected images in the `chosen images` folder, matched each image to its recorded depth using the provided filename-depth spreadsheet, counted white and black pixels, and saved the resulting dataset to the `results` folder as a `.csv` file. The table below is the measured dataset used for the interpolation steps that follow.

[2]:
```
# Now we run the main analysis pipeline, which builds our analysis table and␣
↪saves it to a CSV file.
# The analysis table contains all the relevant data and metrics for our␣
↪fibrosis analysis at the target depth.
analysis_df = main.build_analysis_table()
main.save_analysis_table(analysis_df)

# Finally, we create a display DataFrame that contains only the relevant␣
↪columns for our final results, and we display it in a readable format.
display_df = analysis_df[
    ["basename", "depth_um", "white_pixels", "black_pixels",␣
↪"percent_white_pixels"]
].copy()

# Display the final results in a readable format.
display(display_df)
```

```
Saved analysis table to: E:\OneDrive - University of Virginia\Will
M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_
BME_2315\Module-3-Fibrosis\results\Percent_White_Pixels.csv

                        basename    depth_um  white_pixels  black_pixels  \
0   MASK_Sk658 Llobe ch010036.jpg   55.000000         29677       4164627
1   MASK_SK658 Slobe ch010156.jpg  330.000000         38068       4156236
2   MASK_SK658 Slobe ch010157.jpg  750.000000         56419       4137885
3   MASK_SK658 Slobe ch010159.jpg  860.000000         59426       4134878
4   MASK_SK658 Slobe ch010158.jpg  920.000000         59788       4134516
5   MASK_SK658 Slobe ch010146.jpg 2000.000000         62913       4131391
6   MASK_SK658 Slobe ch010143.jpg 2600.000000         70715       4123589
7   MASK_SK658 Slobe ch010147.jpg 3000.000000         80534       4113770
8   MASK_Sk658 Llobe ch010174.jpg 3100.000000         81014       4113290
9   MASK_SK658 Slobe ch010149.jpg 3350.000000         83416       4110888
10  MASK_Sk658 Llobe ch010034.jpg 6600.000000        101283       4093021
11  MASK_Sk658 Llobe ch010149.jpg 7500.000000        124833       4069471
12  MASK_Sk658 Llobe ch010121.jpg 9000.000000        161111       4033193
13  MASK_SK658 Slobe ch010114.jpg 9900.000000        242340       3951964
14  MASK_SK658 Slobe ch010118.jpg 9900.000000        237289       3957015

    percent_white_pixels
0               0.707555
1               0.907612
2               1.345134
3               1.416826
4               1.425457
5               1.499963
6               1.685977
7               1.920080
```

```
8                1.931524
9                1.988792
10               2.414775
11               2.976251
12               3.841186
13               5.777836
14               5.657411
```

The selected images spanned depths from 55 μm to 9900 μm. Across this subset, the measured fibrosis proxy ranged from **0.707555% white pixels** at **55 μm** to approximately **5.66-5.78% white pixels** at **9900 μm**. These measured values formed the dataset used for plotting and interpolation.

0.13.3 2c. Plot measured fibrosis proxy versus lung depth

Next, we plotted the measured percent white pixels against depth into the lung. This figure shows the measured data only and provides the basis for the interpolation analysis below.

[3]:
```
# Now we can also plot the measured data to visualize our results.
# This will create a plot of the percent of white pixels (which corresponds to␣
↪fibrosis) at the target depth for each sample.
main.plot_measured_data(analysis_df)
```

```
Saved measured-data plot to: E:\OneDrive - University of Virginia\Will
M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_
BME_2315\Module-3-Fibrosis\results\Measured_Percent_White_vs_Depth.png
```

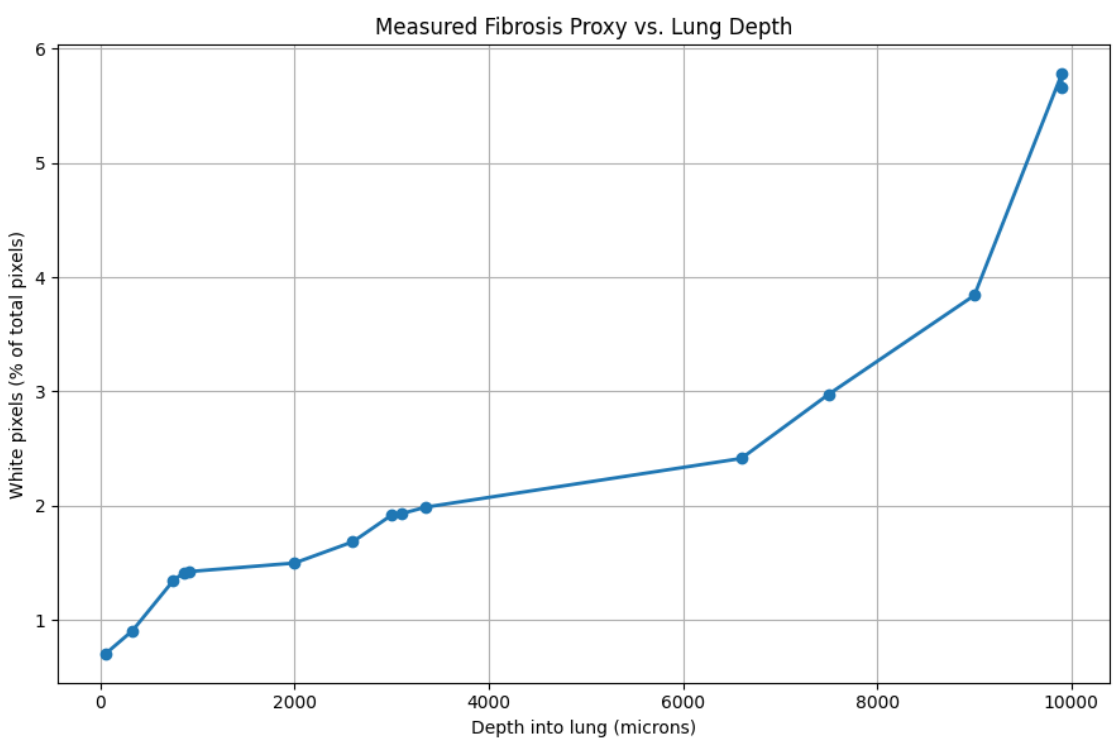


Figure 1. Measured fibrosis proxy versus depth into the lung for the 15 selected images. Each point represents one analyzed image, where the fibrosis proxy is defined as the percent of white pixels after binary thresholding. In this selected subset, we measured percent white pixels generally increased with depth, although local variation remained between neighboring sections and between the two measured images at 9900 μm.

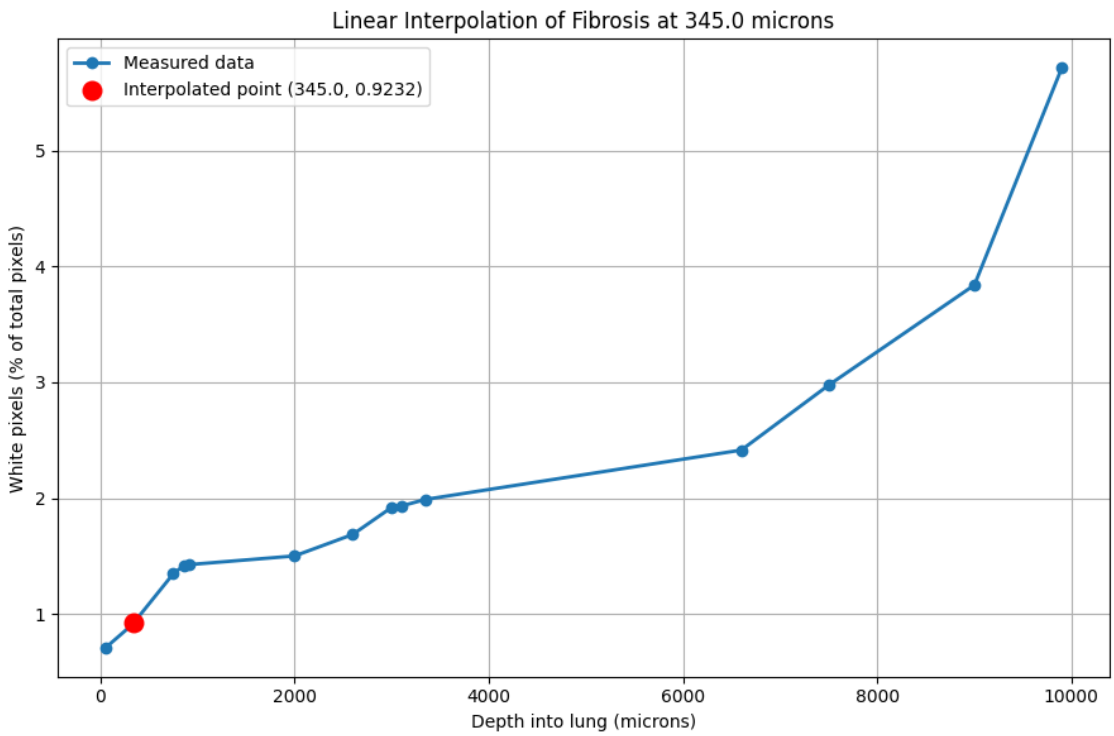
0.13.4 2d. Linear interpolation at 345 μm

After generating the measured dataset, we used linear interpolation to estimate the percent white pixels at **345 μm**, a depth for which we did not directly use one of our selected images. We used both the built-in `interp1d(..., kind='linear')` approach from lecture and a manual matrix-based linear polynomial calculation following the style of `interpolation_example.py`.

[4]:
```
# Next, we run our linear interpolation analysis to estimate the percent of␣
↪white pixels at the target depth using both SciPy's interpolation and our␣
↪manual matrix method.
linear_result = main.run_linear_interpolation(
    analysis_df=analysis_df,
    target_depth_um=TARGET_DEPTH_UM
)

# We then create a summary DataFrame to display these results side by side.
linear_summary = pd.DataFrame(
    {
        "method": ["SciPy linear", "Manual matrix linear"],
        "estimated_percent_white_at_345_um": [
            linear_result["scipy_interpolated_percent_white"],
            linear_result["manual_interpolated_percent_white"],
        ],
    }
)

# Finally, we display the summary of our linear interpolation results.
display(linear_summary)
```

```
Linear interpolation result:
Target depth: 345.0 microns
SciPy linear interpolation: 0.923238% white pixels
Manual matrix linear interpolation: 0.923238% white pixels
Coefficients [a1, a2]: [0.56384461 0.00104172]

Linear interpolation verification:
manual_linear_value: 0.9232376302991594
numpy_linear_value: 0.9232376302991595
absolute_difference: 1.1102230246251565e-16
```

```
Saved linear interpolation plot to: E:\OneDrive - University of
Virginia\Will M\Undergraduate Research Courses-Grants\05_Spring_2026\Computation
al_Biomedical_BME_2315\Module-3-Fibrosis\results\Linear_Interpolation.png
```




```
Saved interpolation results to: E:\OneDrive - University of Virginia\Will
M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_
BME_2315\Module-3-Fibrosis\results\Interpolation_Results.csv
```

```
                 method  estimated_percent_white_at_345_um
0          SciPy linear                            0.923238
1  Manual matrix linear                            0.923238
```

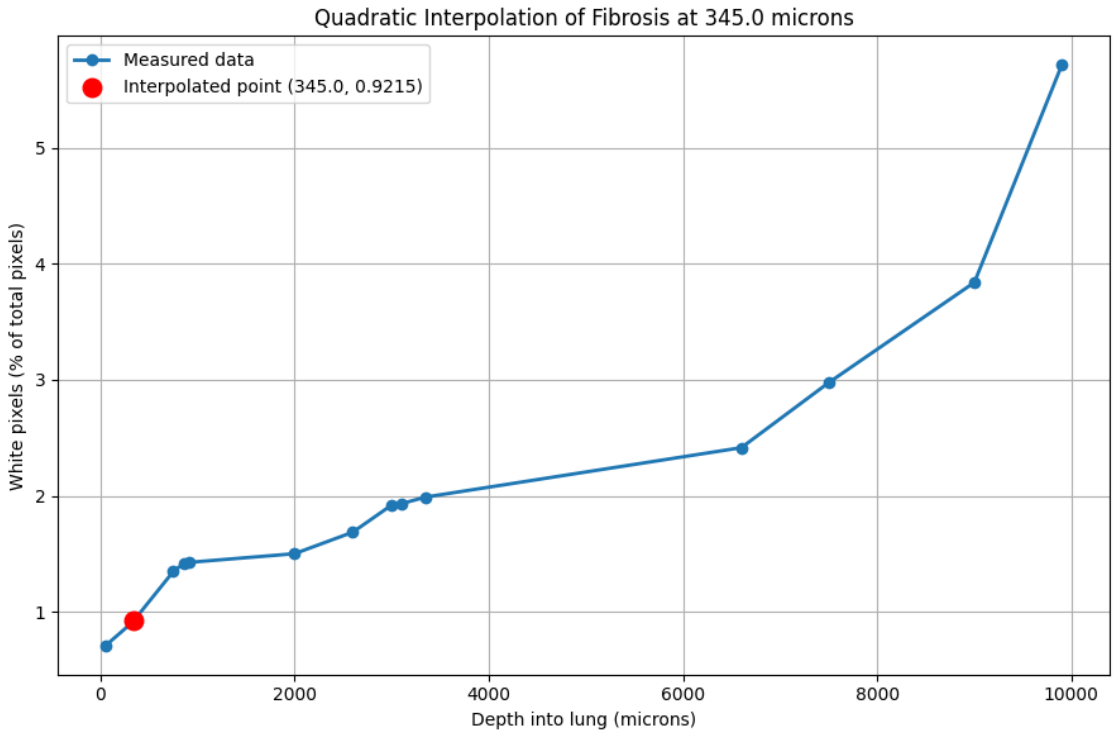
Using linear interpolation, the estimated fibrosis proxy at **345 μm** was **0.923238% white pixels**. The built-in SciPy linear interpolation and the manual matrix-based linear interpolation produced the same value to the displayed precision, showing that both linear implementations were consistent for this target depth.

0.13.5 2e. Quadratic interpolation at 345 μm

We next repeated the estimate using quadratic interpolation. This allowed us to compare a higher-order interpolation method to the linear result and to demonstrate a second interpolation approach, as requested in lecture.

[5]:
```
# Next, we run our quadratic interpolation analysis to estimate the percent of␣
↪white pixels at the target depth using both SciPy's interpolation and our␣
↪manual matrix method.
quadratic_result = main.run_quadratic_interpolation(
    analysis_df=analysis_df,
    target_depth_um=TARGET_DEPTH_UM
)

# We then create a summary DataFrame to display these results side by side.
quadratic_summary = pd.DataFrame(
    {
        "method": ["SciPy quadratic", "Manual matrix quadratic"],
        "estimated_percent_white_at_345_um": [
            quadratic_result["scipy_interpolated_percent_white"],
            quadratic_result["manual_interpolated_percent_white"],
        ],
    }
)

# Finally, we display the summary of our quadratic interpolation results.
display(quadratic_summary)
```

```
Quadratic interpolation result:
Target depth: 345.0 microns
SciPy quadratic interpolation: 0.921540% white pixels
Manual matrix quadratic interpolation: 0.927708% white pixels
Coefficients [a1, a2, a3]: [ 3.81736357e-01  1.83637310e-03 -7.35790932e-07]

Saved quadratic interpolation plot to: E:\OneDrive - University of
Virginia\Will M\Undergraduate Research Courses-Grants\05_Spring_2026\Computation
al_Biomedical_BME_2315\Module-3-Fibrosis\results\Quadratic_Interpolation.png
```




```
Saved interpolation results to: E:\OneDrive - University of Virginia\Will
M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_
BME_2315\Module-3-Fibrosis\results\Interpolation_Results.csv
```

```
                    method  estimated_percent_white_at_345_um
0          SciPy quadratic                            0.921540
1  Manual matrix quadratic                            0.927708
```

Using quadratic interpolation, the built-in SciPy method estimated **0.921540% white pixels** at **345 μm**, while the manual three-point matrix-based quadratic polynomial estimated **0.927708% white pixels**. These values were close but not identical, indicating that the estimated fibrosis proxy at this depth was only mildly sensitive to the interpolation method used.

0.13.6 2f. Summary of interpolation estimates

To summarize the interpolation stage, the table below compares the estimates produced at **345 μm**. Across all methods tested, the predicted fibrosis proxy at this unmeasured depth remained close to **0.92-0.93% white pixels**.

[6]:
```
# To compare all our interpolation results together, we can create a combined␣
↪DataFrame that includes both linear and quadratic interpolation estimates␣
↪from both methods.
interpolation_comparison_df = pd.DataFrame(
    {
```

```
        "method": [
            "SciPy linear",
            "Manual matrix linear",
            "SciPy quadratic",
            "Manual matrix quadratic",
        ],
        "estimated_percent_white_at_345_um": [
            linear_result["scipy_interpolated_percent_white"],
            linear_result["manual_interpolated_percent_white"],
            quadratic_result["scipy_interpolated_percent_white"],
            quadratic_result["manual_interpolated_percent_white"],
        ],
    }
)

# Finally, we display the summary of all our interpolation results together for␣
↪easy comparison.
display(interpolation_comparison_df)
```

```
                    method  estimated_percent_white_at_345_um
0             SciPy linear                            0.923238
1     Manual matrix linear                            0.923238
2          SciPy quadratic                            0.921540
3  Manual matrix quadratic                            0.927708
```

Taken together, the interpolation analysis suggests that the predicted fibrosis proxy at **345 μm** is consistently just under **1% white pixels**. This interpolation step addresses the central project goal of estimating fibrosis at depths where a direct measurement was not part of the selected image subset.

0.13.7 2g. Extra computational benchmark

As an additional computational check, we used the optional runtime benchmark built into `main.py` to estimate how quickly the analysis pipeline could scale to larger image sets. This benchmark was not used to make the biological interpretation, but it helps describe the efficiency of the code implementation.

[7]:
```
# Lastly, we can run our full benchmark pipeline to evaluate the performance of␣
↪our interpolation methods and display the results in a comprehensive␣
↪DataFrame.
benchmark_df = main.benchmark_pipeline()
display(benchmark_df)
```

```
Saved runtime benchmark to: E:\OneDrive - University of Virginia\Will

M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_

BME_2315\Module-3-Fibrosis\results\Runtime_Benchmark.csv
Runtime benchmark:
   n_images_tested  repeats  mean_total_seconds  mean_seconds_per_image  \
```

```
0               15        5            0.130093                0.008673

   estimated_seconds_for_10000_images  estimated_minutes_for_10000_images
0                           86.728867                            1.445481


   n_images_tested  repeats  mean_total_seconds  mean_seconds_per_image  \
0               15        5            0.130093                0.008673

   estimated_seconds_for_10000_images  estimated_minutes_for_10000_images
0                           86.728867                            1.445481
```

For the selected 15 images, the code required about 0.13 seconds per full benchmark run, corresponding to about 0.009 seconds per image. Based on this timing, the same workflow would be expected to process 10,000 images in roughly 1.4-1.5 minutes on this machine.

0.13.8 2h. Verification and Validation Check

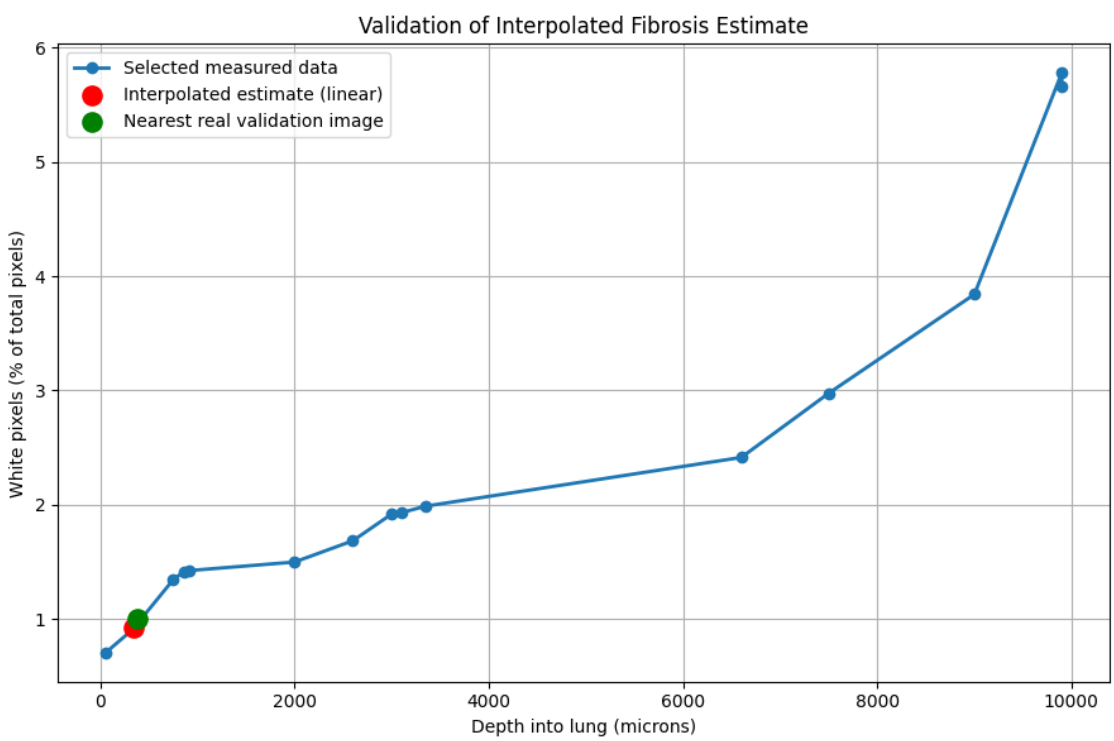
[8]:
```python
# Verification checks on the measured image-analysis table
verification_df = main.verify_analysis_table(analysis_df)
display(verification_df)

# Validation of the interpolation estimate against a nearby real image
validation_df = main.run_validation(
    analysis_df=analysis_df,
    target_depth_um=TARGET_DEPTH_UM,
    method="linear"
)
display(validation_df)
```

```
Verification checks on the image-analysis table:
                        basename  pixel_sum_matches_total  \
0   MASK_Sk658 Llobe ch010036.jpg                     True
1   MASK_SK658 Slobe ch010156.jpg                     True
2   MASK_SK658 Slobe ch010157.jpg                     True
3   MASK_SK658 Slobe ch010159.jpg                     True
4   MASK_SK658 Slobe ch010158.jpg                     True
5   MASK_SK658 Slobe ch010146.jpg                     True
6   MASK_SK658 Slobe ch010143.jpg                     True
7   MASK_SK658 Slobe ch010147.jpg                     True
8   MASK_Sk658 Llobe ch010174.jpg                     True
9   MASK_SK658 Slobe ch010149.jpg                     True
10  MASK_Sk658 Llobe ch010034.jpg                     True
11  MASK_Sk658 Llobe ch010149.jpg                     True
12  MASK_Sk658 Llobe ch010121.jpg                     True
13  MASK_SK658 Slobe ch010114.jpg                     True
14  MASK_SK658 Slobe ch010118.jpg                     True

    percent_between_0_and_100
```

```
0                        True
1                        True
2                        True
3                        True
4                        True
5                        True
6                        True
7                        True
8                        True
9                        True
10                       True
11                       True
12                       True
13                       True
14                       True

                         basename  pixel_sum_matches_total  \
0   MASK_Sk658 Llobe ch010036.jpg                     True
1   MASK_SK658 Slobe ch010156.jpg                     True
2   MASK_SK658 Slobe ch010157.jpg                     True
3   MASK_SK658 Slobe ch010159.jpg                     True
4   MASK_SK658 Slobe ch010158.jpg                     True
5   MASK_SK658 Slobe ch010146.jpg                     True
6   MASK_SK658 Slobe ch010143.jpg                     True
7   MASK_SK658 Slobe ch010147.jpg                     True
8   MASK_Sk658 Llobe ch010174.jpg                     True
9   MASK_SK658 Slobe ch010149.jpg                     True
10  MASK_Sk658 Llobe ch010034.jpg                     True
11  MASK_Sk658 Llobe ch010149.jpg                     True
12  MASK_Sk658 Llobe ch010121.jpg                     True
13  MASK_SK658 Slobe ch010114.jpg                     True
14  MASK_SK658 Slobe ch010118.jpg                     True

    percent_between_0_and_100
0                        True
1                        True
2                        True
3                        True
4                        True
5                        True
6                        True
7                        True
8                        True
9                        True
10                       True
11                       True
12                       True
```

```
13                          True
14                          True
```

```
Saved interpolation validation table to: E:\OneDrive - University of
Virginia\Will M\Undergraduate Research Courses-Grants\05_Spring_2026\Computation
al_Biomedical_BME_2315\Module-3-
Fibrosis\results\Interpolation_Validation.csv
Validation result:
   target_depth_um interpolation_method  predicted_percent_white  \
0       345.000000               linear                 0.923238

       validation_image_basename  validation_image_depth_um  \
0  mask_sk658 llobe ch010040.jpg                 380.000000

   measured_percent_white  absolute_error  percent_relative_error
0                0.999999        0.076761                7.676149

Saved validation plot to: E:\OneDrive - University of Virginia\Will
M\Undergraduate Research Courses-Grants\05_Spring_2026\Computational_Biomedical_
BME_2315\Module-3-Fibrosis\results\Interpolation_Validation.png
```




```
   target_depth_um interpolation_method  predicted_percent_white  \
```

```
0       345.000000               linear                 0.923238

       validation_image_basename  validation_image_depth_um  \
0  mask_sk658 llobe ch010040.jpg                 380.000000

   measured_percent_white  absolute_error  percent_relative_error
0                0.999999        0.076761                7.676149
```

0.14 Verify and validate your analysis:

0.14.1 Verification:

We verified our analysis in a couple of ways. First, we checked the internal consistency of the image-analysis pipeline by confirming that, for every image, the number of white pixels plus the number of black pixels exactly equaled the total number of pixels in that image. We also confirmed that every computed fibrosis proxy value fell within the physically meaningful range of 0% to 100% white pixels. In our selected subset, all 15 images passed both checks, which supports that the thresholding and pixel-counting pipeline was implemented correctly by us.

Second, we verified the interpolation calculation itself by comparing the built-in SciPy linear interpolation result to a manual matrix-based linear interpolation following the polynomial method we did on the whiteboard in class. At the target depth of 345 μm, both approaches gave the same estimate to displayed precision:

$$0.923238\% \text{ white pixels.}$$

The absolute difference between the manual linear interpolation value and NumPy's linear interpolation check was

$$1.1102230246251565 \times 10^{-16},$$

which is effectively zero and indicates that the linear interpolation implementation was numerically consistent.

We also compared interpolation approaches more broadly. The four estimates at 345 μm were:

- SciPy linear: **0.923238%**
- Manual matrix linear: **0.923238%**
- SciPy quadratic: **0.921540%**
- Manual matrix quadratic: **0.927708%**

Because all four methods produced values in a narrow range of approximately **0.92-0.93% white pixels**, our estimate at the target depth appears stable and not overly sensitive to the interpolation method chosen.

0.14.2 Validation:

To validate our interpolation result, we compared the estimated fibrosis proxy at 345 μm to a real image from the full dataset located near that same depth. Using the linear interpolation result from our selected subset, the predicted fibrosis proxy at 345 μm was

$$0.923238\% \text{ white pixels.}$$

The nearest real validation image outside the selected subset was `mask_sk658 llobe ch010040.jpg`, obtained at a depth of **380 μm**. When we analyzed that actual image, we measured

$$0.999999\% \text{ white pixels.}$$

The absolute error between the interpolated prediction and the measured validation image was therefore

$$|0.999999 - 0.923238| = 0.076761 \text{ percentage points,}$$

and the percent relative error was

$$\frac{0.076761}{0.999999} \times 100\% \approx 7.676149\%.$$

This is a relatively small error, especially given that the validation image was not taken at exactly 345 μm but rather at 380 μm. This comparison supports that our interpolation-based estimate is reasonable and that the pipeline can provide a useful approximation of fibrosis severity at depths where a direct image measurement is not available.

Overall, our verification checks support that the code is internally consistent, and our validation step supports that the interpolation result is biologically plausible when compared to a nearby real measurement.

0.15 Conclusions and Ethical Implications:

0.15.1 Conclusions:

Our analysis suggests that the fibrosis proxy, measured as percent white pixels in black-and-white lung sections, generally increased with depth in our selected subset of images. In the measured dataset, the fibrosis proxy ranged from **0.707555% white pixels** at **55 μm** to approximately **5.66-5.78% white pixels** at **9900 μm**. Although there was some local variation between nearby sections, the overall trend indicates that fibrosis was not distributed uniformly throughout the sampled depth range.

Using interpolation, we estimated the fibrosis proxy at a depth of **345 μm**, where we did not directly include one of our selected images. All interpolation methods produced a very similar result, with predicted values ranging from approximately **0.921540%** to **0.927708% white pixels**. Taken together, these results suggest that the fibrosis burden at 345 μm is just under **1% white pixels**.

In the context of our project goal, these findings support the idea that a computational image-analysis pipeline can be used not only to quantify fibrosis at measured biopsy depths, but also to estimate fibrosis at nearby unmeasured depths. This is useful because, in a real biopsy setting, it is not practical to collect tissue at every possible depth throughout the lung. A depth-aware interpolation framework can therefore help predict how fibrosis changes through tissue and can support the design of a biopsy device that targets regions more precisely.

Our results also suggest an important design implication for such a device: because fibrosis appears to vary with depth rather than remaining constant, a next-generation biopsy tool would benefit from accurate depth control and depth tracking. If fibrosis severity changes meaningfully over short distances, then sampling from the wrong depth could underrepresent or overrepresent the true extent of disease. Therefore, based on our analysis, we recommend that a future lung biopsy device incorporate accurate depth targeting and depth tracking so that tissue can be sampled from clinically meaningful regions of fibrosis.

0.15.2 Ethical Implications:

Although this analysis was useful, the predictions generated by the interpolation model should not be treated as exact science or truth. Interpolation is an estimate based on neighboring data points, and real tissue is heterogeneous. A predicted fibrosis value at an unmeasured depth could differ from the true biological value if the tissue changes abruptly or if the selected images do not fully represent the surrounding lung region. For this reason, interpolation should support medical decision-making, not replace direct measurement, pathology review, or clinical judgment.

There are also ethical considerations related to translation from a mouse model to human disease. Our dataset came from a fibrotic mouse lung (commonly used in research here at UVA), which is a valuable research model but not a perfect representation of human lung fibrosis. A device or algorithm developed from this kind of dataset must therefore be validated carefully before being applied clinically. Overgeneralizing from animal data to human patients could lead to misleading conclusions or inappropriate clinical decisions.

Another ethical issue is sampling bias. If a biopsy device samples tissue from only one location or one depth, it may miss other regions with different fibrosis severity. Since our results suggest that fibrosis is depth-dependent and locally variable, inaccurate or poorly targeted sampling could cause clinicians to underestimate disease severity in one patient or overestimate it in another. This reinforces the importance of precise device placement and transparent communication about model uncertainty to the stakeholders involved (patients).

Finally, if depth-specific biopsy technologies become clinically useful, they should be implemented in a way that promotes equitable access for everyone. Advanced image-guided or depth-guided biopsy tools could improve diagnosis, but only if they are available to a broad range of patients rather than being limited to well-resourced medical centers. Ethical development of such technology should therefore consider not only technical accuracy, but also safety, transparency, and fairness in who benefits from the device. This is all aligned with the goal of providing quality medical science to ALL of those who need it.

0.16 Limitations and Future Work:

One important limitation of this project is that our analysis used only a **selected subset of 15 images** from the larger 78-image class dataset. Although this subset was sufficient to demonstrate the image-analysis and interpolation workflow, it may not fully capture the spatial heterogeneity of fibrosis throughout the entire lung. A second limitation is that our fibrosis proxy was based on **binary thresholding** of black-and-white images. This is computationally simple and reproducible, but it reduces a complex histologic pattern to a single summary metric and may miss subtler differences in lesion morphology, intensity, or tissue architecture.

Another limitation is that interpolation assumes that fibrosis changes relatively smoothly between

nearby depths. In real tissue, however, fibrosis may vary abruptly over short distances because lung scarring is spatially heterogeneous. For this reason, an interpolated value should be interpreted as an estimate rather than an exact biological truth. Our validation result was encouraging, but it was still based on a nearby image at **380 μm** rather than an image taken at exactly **345 μm**. In addition, this dataset came from a **murine bleomycin model**, which is useful for studying fibrotic mechanisms but does not perfectly reproduce the full biology or clinical variability of human idiopathic pulmonary fibrosis [8].

If we had more time, the first improvement would be to analyze **all 78 provided images** rather than only a selected subset. That would allow denser sampling across lung depth and would likely improve interpolation accuracy. We would also like to test additional computational approaches, such as cubic interpolation, spline fitting, adaptive thresholding, or more advanced segmentation methods, to see whether they better capture local changes in fibrosis. A stronger validation framework would include holding out multiple images at different depths and comparing predicted values against measured values across the full lung rather than at only one target depth.

For future work related to the device-design goal, we would also like to connect the image-analysis pipeline more directly to biopsy-device development. For example, a future study could identify depth ranges with the highest change in fibrosis burden and use that information to define target sampling windows for a depth-aware biopsy device. Ultimately, integrating computational fibrosis prediction with precise biopsy-depth control could help produce more informative tissue sampling and reduce the risk of underestimating or overestimating disease severity.

0.17 Notes:

- **Completed:**
  - Data and Disease Background: Roy
  - Data Analysis: Will and Roy
  - Verification/Validation: Will
  - Conclusions/Ethics: Will
  - Limitations/Future Work: Roy
- **To Do:**
  - Nothing at this time.
- **Questions/Concerns:**
  - Nothing at this time.
- **References:**

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  - [7] University of Virginia Health, "John Kim, MD | Pulmonary and Critical Care," UVA

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  - [8] BME 2315, "Module 3 Lecture 1: Fibrosis," lecture materials, University of Virginia, 2026.
  - [9] "Filenames and Depths for Students.csv," provided lookup table, BME 2315, University of Virginia, 2026.

- **AI Statement:**
  - External use of LLMs was utilized on a case-by-case basis, often for initial source discovery + inspiration and code debugging. GitHub Copilot (which we students have a free trial) was especially helpful with providing 'Inline Suggestions' for shortening time with boilerplate code and repetitive function naming. This was particularly helpful for implementing LaTeX format into our markdown segments to that we can easily see the equations being used in a clear sense. However, the final code structure, modeling choices, and interpretations were based on the course lectures, GitHub materials, and provided resources, with all final analysis completed by our team.